Name: Joseph Everest Lyimo

Student ID: 618981

Project Name: ShelfLife

Description: A store and inventory management system

1. Introduction

Many general retail stores begin as small operations where a single owner can track stock, sales, and expenses in a notebook or a simple spreadsheet. When the store carries only a handful of products and serves a predictable stream of customers, this manual approach works well enough, and the owner can keep the whole picture in their head.

As a store grows, this breaks down. It now carries hundreds or thousands of distinct products (SKUs), employs sales clerks and stock keepers, buys from several suppliers on different credit terms, and extends credit to regular customers. Under that load, stock counts drift out of sync with reality, fast-moving items run out without warning, and profit becomes hard to compute once goods are bought in bulk but sold as single units. Debts owed to suppliers and money owed by customers end up tracked on scraps of paper, and no single person has a reliable, up-to-date picture of the store's financial position.

A general store also has natural divisions of work. An owner sets the business up; a manager sets prices and orders stock; clerks sell to customers or receive and count deliveries; and an accountant keeps the books for tax. Each of these needs a different level of access and a different view of the same underlying data.

ShelfLife is a new software tool that manages the full cycle of a general store's operations: it tracks inventory from bulk purchase through to unit sale, records sales at manager-controlled prices, manages purchase orders and supplier payments, invoices customers who buy on credit, records operating expenses, and computes profit per item all under a role-based permission system so each employee sees and does only what their role allows. Simple stock projections are included as a convenience feature to help anticipate restocking.

2. Positioning

2.1 Problem Statement

The problem of managing stock, sales, purchasing, and store finances by hand as a general store grows,

affects store owners, managers, sales and inventory clerks, and accountants, the impact of which is unexpected stock-outs and lost sales, inaccurate or unknown profit, supplier and customer debts that are easily lost track of, and the absence of a single reliable picture of the store's financial health, a successful solution would be one integrated tool that tracks inventory from bulk purchase to unit sale, records sales at controlled prices, manages suppliers and purchase orders with flexible payment terms, invoices credit customers, records expenses, and calculates profit, all behind a permission system so that each role does only what it should. This tool provides a shared database and a user interface that is easy to use for owners, managers, clerks, and accountants.

2.2 Product Position Statement

For the owner, managers, and staff of a growing general retail store who need to keep stock, sales, purchasing, and finances accurate and under control as the business outgrows notebooks and spreadsheets, ShelfLife is a general store management system that tracks inventory from bulk purchase to unit sale, controls pricing, manages suppliers and credit customers with flexible payments, records expenses, and reports profit — all under role-based access.

Unlike manual spreadsheets or a generic point-of-sale application, our product integrates inventory, sales, purchasing, supplier and customer credit, and expenses in one place, with a bulk-to-unit stock model and velocity-based low-stock alerts tailored to how a general store actually operates.

3. Stakeholder Descriptions

3.1 Stakeholder Summary

Name	Description	Responsibilities
Admin / Owner	The person who sets up the store in the system for the first time. Holds root-level permissions.	Names and configures the store, invites employees and assigns their roles, defines product categories, and can view all analytics and reports.
Manager	Runs day-to-day operations. May be blended with the Admin/Owner role in a small store.	Sets and edits selling prices, creates purchase orders to suppliers, approves received stock, and reviews sales, stock, and profit analytics.
Sales Clerk	Front-of-store staff who serves customers at the point of sale.	Records sales at the prices set by the Manager, and raises credit invoices for customers buying on account.
Inventory Clerk	Back-of-store staff who manages stock. A Clerk may handle sales, inventory, or both.	Adds products and stock in bulk/SKU quantities, defines the SKU-to-unit breakdown, and records and signs off deliveries after verification. Read/write only — cannot edit records after they are published.
Accountant	A finance role focused on the store's books and tax obligations.	Records operating expenses, and views financial reports (revenue, cost of goods, expenses, and profit) for tax purposes.
Developers	Build the system based on this document and the models derived from it.	Develop system features, fix defects, and maintain the availability of the system.
Testers	Verify the system behaves as specified.	Perform unit and integration testing (including the business rules for stock, pricing, and payments).

3.2 User Environment

The system is used by the staff of a single general store. A small store may have as few as two or three people (an owner who also manages, a clerk, and an accountant); a larger store may have a manager, several sales and inventory clerks working across shifts, and a dedicated accountant. The Admin/Owner and Manager roles may be held by the same person.

Work happens in two main settings: at the sales counter, where clerks record sales and take payment, often quickly and repeatedly through the day; and in the stockroom, where inventory clerks receive deliveries, count and verify goods against purchase orders, and sign them into stock. Managers and the accountant work more periodically on setting prices, placing orders, recording expenses, and reviewing reports.

The application is a multi-user, multi-tier web application, so it is reached through a web browser on a computer, tablet, or phone connected to the store's network or the internet. It is expected to integrate with no external system in its first release; it is self-contained, with all data held in the store's own database. (Barcode scanning and receipt printing are natural future integrations but are not required for the first release.)

4. Product Overview

4.1 Product Perspective

ShelfLife is an independent, self-contained system. It is organized into cooperating subsystems that share a single database:

- Authentication & Permissions: sign-in and role-based access control across the whole system.
- Inventory: products/SKUs, the bulk-to-unit breakdown, stock levels, and velocity-based low-stock alerts.
- Sales: recording sales at manager-set prices and computing per-item profit.
- Purchasing & Suppliers:suppliers, purchase orders, delivery sign-off, and supplier payment (immediate, deferred, or partial).
- Customer Invoicing: credit sales and customer balances.
- Expenses & Accounting: operating expenses and financial reporting for tax.
- Analytics & Projections: dashboards and simple stock-keeping projections.

The application follows a multi-tier architecture: a web presentation tier (the user interface), an application tier holding the business logic for the subsystems above, and a data tier (a relational database). A block diagram showing these subsystems, the shared database, and the browser clients can be added here.

4.2 Assumptions and Dependencies

- The system manages a single store; multiple branches are out of scope for the first release.
- All monetary values are in a single currency configured at setup.
- Users have a modern web browser and network/internet access to the application.
- The Admin/Owner performs initial setup and creates the first accounts before other roles use the system.
- Products are bought in consistent bulk units that can be defined as a fixed number of sale units (e.g. 1 dozen = 12 units).
- Demand forecasting beyond simple stock projections (full ERP-style analytics) is out of scope; if that changes, this document must be revised.

4.3 Needs and Features

The following table lists the problems the product addresses, the need behind each, a priority, the corresponding features (described at a general capability level, not design), and the planned release.

No	Problem	Need	Priority	Features	Planned Release
Setup & Administration (Admin / Owner)					
1	A new store must be set up before it can be used.	The store must be created and named in the system.	High	Admin can create and name the store and set basic details such as currency and address.	1.0
2	Staff need access to the system with the right level of rights.	Employees must be invited and given roles.	High	Admin/Manager can invite employees and assign roles (Manager, Sales Clerk, Inventory Clerk, Accountant).	1.0
3	Products need to be organized so they can be found and reported on.	Product categories must exist.	Medium	Admin/Manager can add, edit, or delete product categories.	1.0

No	Problem	Need	Priority	Features	Planned Release
4	Different roles must not see or change everything.	Access must be controlled per role.	High	The system enforces role-based permissions across all subsystems; clerks are limited to read/write with no editing after publish.	1.0
Inventory					
5	The store carries many distinct products.	Products / SKUs must be created in the system.	High	Inventory Clerk/Manager can add a product with its SKU, category, and unit-of-sale definition.	1.0
6	Stock is bought in bulk but sold in single units.	Bulk quantities must be broken down into sellable units.	High	The system converts a bulk/SKU quantity into sale units using a defined breakdown (e.g. 1 dozen = 12 units); the clerk defines what those units are.	1.0
7	Stock counts drift out of sync when tracked by hand.	Stock levels must always reflect purchases and sales.	High	The system automatically increases stock when goods are received and decreases it when a sale is recorded.	1.0
8	Records could be quietly altered, harming accuracy and audit trails.	Published records must be protected.	Medium	Clerks have read/write access but cannot edit a record once it has been published, preserving audit integrity.	1.0
9	Fast-moving items run out unexpectedly.	Low stock must be flagged based on how fast an item actually sells.	High	The system flags a product as "running low" using its historical sales velocity against remaining stock (e.g. 200 left of an item selling 100/day is low; 50 left of an item selling 50/month is not urgent).	1.0
Sales					
10	Customers buy products and sales must be recorded.	Clerks must be able to process a sale.	High	Sales Clerk can record a sale at the Manager-set price; stock decrements automatically.	1.0
11	Prices must stay under management control.	Only Managers may set prices.	High	Manager sets and edits selling prices; clerks sell at those prices and cannot change them.	1.0
12	The store needs to know its profit on each item.	Per-item profit must be computed.	High	The system computes per-item profit from bulk cost versus unit selling price (e.g. a dozen bought for 5,000 and sold as 12 units at 500 = 6,000, a profit of 1,000).	1.0
Purchasing & Suppliers					

No	Problem	Need	Priority	Features	Planned Release
13	The store restocks from a number of suppliers.	Supplier records must be maintained.	High	Manager can add, edit, or delete suppliers and track each supplier's details and outstanding balance.	1.0
14	Restocking must be requested in an organized way.	Purchase orders must be created.	High	Manager creates purchase orders to suppliers.	1.0
15	Deliveries must be checked before they enter stock.	Received goods must be verified and signed off.	High	Inventory Clerk/Manager records a delivery, verifies it against the purchase order, and signs off; stock updates only on sign-off.	1.0
16	Suppliers are sometimes paid on a future day, not on delivery.	Supplier payment terms must be flexible.	High	On receipt the user may pay immediately, defer payment, or make a partial payment; the system tracks each supplier's outstanding balance.	1.0
Customer Invoicing (Credit Sales)					
17	Some customers buy on credit rather than paying at once.	Credit sales must be tracked per customer.	Medium	The system records a credit sale as a customer invoice and tracks each customer's outstanding balance.	1.0
18	Credit is settled over time, sometimes in parts.	Customer payments must be recorded against invoices.	Medium	The system records full or partial payments against a customer invoice and updates the balance.	1.0
Expenses & Accounting					
19	Running the store incurs ongoing costs.	Operating expenses must be recorded.	Medium	Accountant/Manager records expenses by category: rent, transport, delivery, employee salaries, and utilities.	1.0
20	The store must report its finances, including for tax.	Financial data must be summarized.	Medium	Accountant can view financial reports — revenue, cost of goods sold, expenses, and profit — for tax purposes.	1.0
Analytics & Projections					
21	The owner needs visibility into how the store is doing.	Analytics must be available to authorized roles.	Medium	Admin/Manager views analytics for sales, top products, profit, and stock status.	1.0
22	The store should anticipate when to restock.	Simple stock projections are helpful (fuller forecasting leans toward ERP and is out of scope).	Low	The system provides simple stock-keeping projections, such as an estimated number of days until an item runs out, from its average sales rate.	2.0

4.4 Alternatives and Competition

Stakeholders perceive several alternatives to building ShelfLife:

- Status quo, notebooks and spreadsheets. Strength: free, familiar, no setup. Weakness: does not scale, error-prone, no shared real-time view, no role control, and no automatic profit or stock tracking.
- Off-the-shelf point-of-sale apps (e.g. Loyverse, Square). Strength: quick to adopt, good at the checkout. Weakness: often weaker on bulk-to-unit stock modelling, supplier credit, and customer invoicing tailored to a general store; may carry ongoing per-terminal costs.
- Full ERP systems (e.g. Odoo, SAP Business One). Strength: very capable and comprehensive. Weakness: costly, complex, and far heavier than a single general store needs, much of the capability would go unused.

ShelfLife aims to sit between the last two: purpose-built for a general store's inventory, purchasing, credit, and expense needs, without the cost and complexity of a full ERP.

5. Other Product Requirements

- Platform: a web application with a relational database backend. Built in React and Typescript with MySQL as the database
- Security: all access is authenticated; role-based permissions are enforced on every subsystem; published records cannot be edited by clerks, preserving an audit trail.
- Usability: the interface must be simple enough for counter staff to record sales quickly with minimal training.
- Reliability & data integrity: stock, sales, and payment records must remain consistent (e.g. stock only changes through recorded sales and signed-off deliveries).
- Performance: routine actions such as recording a sale or looking up stock should respond quickly under normal store load.
- Documentation: basic user guidance for each role (Admin/Owner, Manager, Clerk, Accountant) should be provided.
- Constraints: single store, single currency, and no required external integrations in the first release (barcode scanning and receipt printing are candidate future enhancements).